

Sri Lanka A/L Combined Mathematics

Master Limit Question Bank: 50 Structured Problems

Aligned with the Sri Lankan G.C.E. Advanced Level Syllabus (Calculus: Limits)

Instructions

This question paper contains 50 carefully curated limit problems modeled directly on past Sri Lankan G.C.E. Advanced Level Combined Mathematics (Calculus) examinations. Use standard limit theorems, trigonometric identities, rationalization, and factorization techniques to evaluate each limit.

1. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x}$
2. Evaluate $\lim_{x \rightarrow 0} \frac{\tan 4x}{x}$
3. Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$
4. Evaluate $\lim_{x \rightarrow 0} \frac{\sin^2 3x}{x \tan 2x}$
5. [Reference Style] Show that $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x(\sqrt{3x^2 + 1} - \sqrt{x^2 + 1})} = \frac{1}{2}$
6. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x - \sin 3x}{\sin x}$
7. Evaluate $\lim_{x \rightarrow 0} \frac{\cos 3x - \cos x}{x^2}$
8. Evaluate $\lim_{x \rightarrow 0} \frac{\tan 3x - \sin 3x}{x^3}$
9. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$
10. Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$
11. Evaluate $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$
12. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x \sin 3x}{x^2}$
13. Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x \sin 2x}$
14. Evaluate $\lim_{x \rightarrow 0} \frac{\tan 2x - \sin 2x}{x^3}$
15. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{4+x} - 2}{x}$
16. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+2x} - \sqrt{1-2x}}{x}$
17. Evaluate $\lim_{x \rightarrow 0} \frac{\tan^2 x}{1 - \cos x}$

18. Evaluate $\lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x}$
19. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{\sqrt{1+x} - 1}$
20. Evaluate $\lim_{x \rightarrow 0} \frac{x^2}{\cos x - \cos 3x}$
21. Evaluate $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$
22. Evaluate $\lim_{x \rightarrow 0} \frac{\tan(\sin x) - \sin x}{x^3}$
23. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1}{x^2}$
24. Evaluate $\lim_{x \rightarrow 0} \frac{\sin(a+x) - \sin(a-x)}{x}$
25. Evaluate $\lim_{x \rightarrow 0} \frac{\cos(a+x) - \cos(a-x)}{x}$
26. Evaluate $\lim_{x \rightarrow \pi/2} \frac{\cos x}{\frac{\pi}{2} - x}$
27. Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$
28. Evaluate $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x}$
29. Evaluate $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$
30. Evaluate $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$
31. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x + \sin 7x}{x}$
32. Evaluate $\lim_{x \rightarrow 0} \frac{\tan 5x - \tan 3x}{x}$
33. Evaluate $\lim_{x \rightarrow 0} \frac{x \tan x}{1 - \cos x}$
34. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{9+x} - 3}{x}$
35. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x^2+9} - 3}{\sqrt{x^2+16} - 4}$
36. Evaluate $\lim_{x \rightarrow 0} \frac{\sin^3 x}{x \sin x^2}$
37. Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin 2x}$
38. Evaluate $\lim_{x \rightarrow 0} \frac{\sec x - 1}{x^2}$
39. Evaluate $\lim_{x \rightarrow 0} \frac{\csc x - \cot x}{x}$

40. Evaluate $\lim_{x \rightarrow 0} \frac{\tan 3x - 3 \tan x}{x^3}$

41. Evaluate $\lim_{x \rightarrow 0} \frac{\ln(1 + 3x)}{x}$

42. Evaluate $\lim_{x \rightarrow 0} \frac{e^{3x} - e^x}{x}$

43. Evaluate $\lim_{x \rightarrow 0} \frac{x(\sqrt{x^2 + 4} - 2)}{\sin^2 x}$

44. Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$

45. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{2 - \cos x} - 1}{\sin^2 x}$

46. Evaluate $\lim_{x \rightarrow 0} \frac{x^2 \sin(\frac{1}{x})}{\sin x}$

47. Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - x}{x - \sin x}$

48. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1 + x^3} - 1}{x^3}$

49. Evaluate $\lim_{x \rightarrow 0} \frac{x^3 \cot x}{1 - \cos x}$

50. Evaluate $\lim_{x \rightarrow 0} \frac{\tan(\frac{\pi}{4} + x) - \tan(\frac{\pi}{4} - x)}{x}$

End of Question Paper